

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Farmer Input Management	Enter soil and environmental parameters (Nitrogen, Phosphorus, Potassium, pH, Temperature, Humidity, Rainfall).
FR-2	User Input Validation	Validate all input fields, ensure numeric values are entered, check for missing or invalid data, and display appropriate error messages.
FR-3	Data Processing & Preprocessing	Prepare the validated input data, organize feature values, and convert them into the format required by the machine learning model.
FR-4	Machine Learning Model Management	Load the trained machine learning model and label encoder during application startup to enable crop prediction.
FR-5	Crop Recommendation Engine	Process the validated input data using the trained machine learning model and generate the most suitable crop recommendation.
FR-6	Prediction Result Display	Display the recommended crop clearly on the web interface and allow the user to perform another prediction.
FR-7	Error Handling	Handle invalid inputs, unexpected system errors, and model prediction failures by displaying meaningful error messages without crashing the application.
FR-8	Agricultural Decision Support	Assist farmers in selecting the most suitable crop based on soil nutrients and environmental conditions to support informed agricultural decisions.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple, intuitive, and user-friendly interface that can be easily used by farmers and agricultural stakeholders.

NFR-2	Security	Validate all user inputs and securely process user-provided agricultural data without unauthorized modification or exposure.
NFR-3	Reliability	Produce accurate and consistent crop recommendations whenever valid input data is provided.
NFR-4	Performance	Generate crop prediction results in less than 2 seconds under normal operating conditions.
NFR-5	Availability	Ensure the application remains available whenever the Flask server is running.
NFR-6	Scalability	Support future enhancements, including additional crop datasets, larger user bases, and integration with advanced agricultural services.
NFR-7	Maintainability	Use a modular software architecture that allows easy updates, debugging, and future feature additions.
NFR-8	Accuracy	Generate highly accurate crop recommendations using the trained machine learning model and validated agricultural data.
NFR-9	Portability	Allow the application to run on different operating systems and be accessible through modern web browsers.
NFR-10	Data Privacy	Ensure secure handling and temporary processing of user-provided agricultural data without unnecessary storage.
NFR-11	Compatibility	Ensure compatibility with modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.